

DESIGN OF CIRCULAR THIN-WALLED CONCRETE FILLED STEEL TUBES

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ABSTRACT: In this paper several design methods have been developed that can be used to conservatively estimate the strength of circular thin-walled concrete filled steel tubes under different loading conditions. The loading conditions examined include axial loading of the steel only, axial loading of the concrete only, and simultaneous loading of the concrete and steel both axially and at small eccentricities. Recent tests on circular concrete filled steel tubes have been used to calibrate and validate the proposed design methods. The test specimens were short with a length-to-diameter ratio of 3.5 and a diameter thickness ratio between 60 and 220. The internal concrete had nominal unconfined cylinder strengths of 50, 80, and 120 MPa. The bond (or lack of) between the steel and internal concrete was critical in determining the formation of a local buckle.

INTRODUCTION

Concrete filled steel tubes are an economical column type as the majority of the axial load is resisted by the concrete, which is less expensive than steel. Further economies can be obtained using high-strength concrete with thin-walled steel tubes, using just sufficient steel to support the construction loads prior to filling with concrete.

At the present time, there is no design standard that can be used for the design of thin-walled steel tubes filled with concrete. For thick-walled steel tubes filled with normal-strength concrete, some design guidance is given in codes such as Eurocode 4 [European Committee for Standardization (CEN) 1992] and ACI 318 [American Concrete Institute (ACI) 1989]. These two methods are quite different in concept. Eurocode 4 (CEN 1992) uses a column curve to determine the influence of slenderness (similar to most modern steel codes). Additionally, enhancement of the concrete from confinement is included for some specific cases. The ACI 318 (ACI 1989) method used the traditional reinforced concrete approach, similar to AS3600 [Standards Association of Australia (SAA) 1994] with a minimum load eccentricity used to determine the column strength under nominal axial load. Bridge (1990) proposed a variation on the reinforced concrete approach, similar to the method in AS3600 (AS 1994).

An extensive series of tests have been conducted by O'Shea and Bridge (1997a,b) examining the influence of thin-walled steel tubes and very-high-strength concrete on the strength of short circular concrete filled steel tubes. From these results, different design methods have been evaluated and recommended to predict the strength of this type of member under a variety of loading conditions.

PAST RESEARCH

A significant amount of effort over the past 40 years has been aimed at developing a better understanding of circular concrete filled steel tubes. Literature reviews by O'Shea and Bridge (1997a,b) indicated that thick-walled concrete filled steel tubes have received extensive attention by a number of authors (Boué 1957; Klöppel and Goder 1957; Salani and Sims 1964; Furlong 1967; Gardner and Jacobson 1967; Gard-

ner 1968; Knowles and Park 1969, 1970; Neogi et al. 1969; Guiaux and Janss 1970; Sen 1972; Lin 1988; Roik and Bergman 1989; Cai 1991; Masuo et al. 1991; Sakino and Hayashi 1991; Imamura et al. 1994). By comparison, only limited attention has been paid to either circular thin-walled steel tubes with medium-strength concrete or circular thick-walled steel tubes with high-strength concrete (Prion and Boehme 1989; University of Sydney 1990; Luksha and Nesterovich 1991; Rangan and Joyce 1992; Kavoossi and Schmidt 1993; Kilpatrick 1996) with no author examining circular thin-walled steel tubes filled with high strength concrete.

Design guidelines for circular thick-walled steel tubes filled with normal-strength concrete were initially proposed by Dowling et al. (1977). The influences of concrete enhancement, steel reduction (due to biaxial effects), and column slenderness were all incorporated. This work and that by Roik and Bergmann (1989) formed the basis for the design rules in Eurocode 4 (CEN 1992) for circular concrete filled steel tubes.

The influence of internal restraint on the buckling behavior of thin-walled tubes has been examined by Grimault and Janss (1977). The available test results indicated an enhancement of the local buckling strength of the steel tube from the internal concrete. Consequently, an equation was developed to give a better estimate of the ultimate strength of the thin-walled steel tube based upon an effective area approach. Later tests by O'Shea and Bridge (1997a,b,c) however have indicated that the improved strength of the circular thin-walled tubes, filled with concrete, is due to the influence of bond between the steel tube and concrete rather than lateral restraint.

AXIAL STRENGTH

As concrete filled steel tubes are generally filled on-site after erection, an accurate estimate of the unfilled strength is important. This is especially the case for circular thin-walled steel tubes under high construction loads. Issues on the service strength of the steel tube also need to be addressed as the concrete infill may provide lateral restraint to the steel tube, restraining the formation of a local buckle and forcing a different buckling mode, thereby increasing the column's capacity.

Circular Steel Tubes, Steel Loaded

The strength of thin-walled circular steel tubes can be significantly influenced by local buckling. In Eurocode 4 (CEN 1992) the influence of local buckling has been considered by restricting the allowable diameter to thickness ratio. Unlike other design codes, which have similar restrictions, there is little guidance for tubes beyond this limit, except specifying that an experimentally confirmed method has to be used.

The behavior of steel loaded, thin-walled circular steel tubes with and without internal restraint has been experimentally

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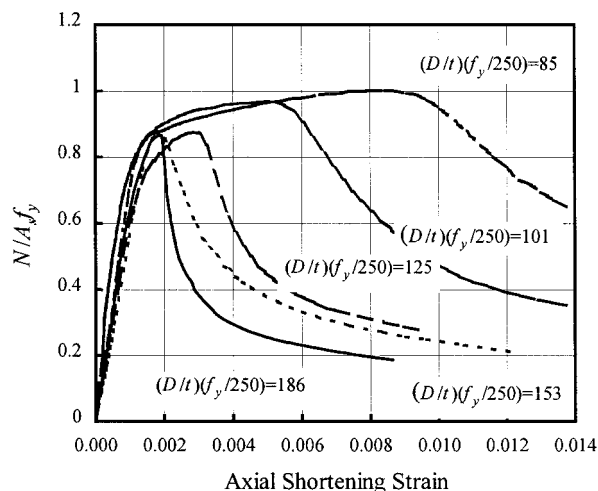


FIG. 1. Normalized Response for Axially Loaded Unfilled Circular Steel Tubes

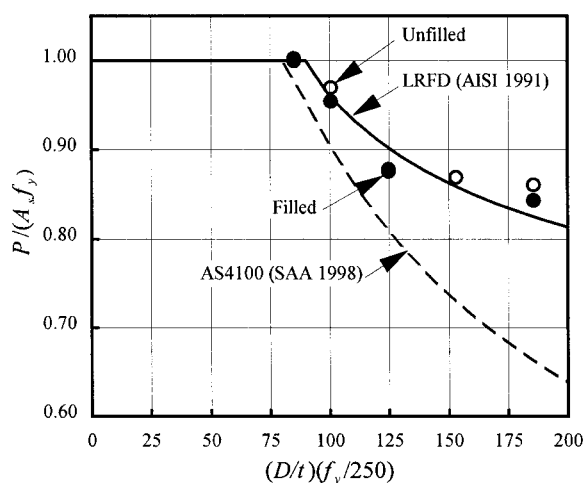


FIG. 2. Comparison of Circular Steel Tubes with AS4100 (SAA 1990) and LRFD (AISI 1991)

investigated by O'Shea and Bridge (1997c). The specimens examined were short ($L/D = 3.5$) with a range of diameter-to-thickness ratios. The load-axial shortening response of the unfilled specimens (normalized with respect to the yield load on the steel) are plotted in Fig. 1. Evidently, the strength and ductility decrease with increasing diameter-to-thickness ratios. Further tests were conducted to examine the influence of internal restraint by filling with concrete. The ultimate load of the filled and unfilled specimens, normalized with respect to the yield load on the steel are plotted against the diameter-to-thickness ratio in Fig. 2. The concrete has had little effect on the ultimate load, as the predominantly outward buckle was not restrained by the concrete infill.

As the thin-walled circular steel tubes have the same ultimate axial load, whether filled or unfilled with concrete, O'Shea and Bridge (1997c) proposed that existing standards for unfilled steel tubes can be used to calculate the strength of thin-walled circular steel tubes with concrete infill. A good strength estimate (ignoring ϕ factors) can be obtained from the LRFD [American Iron and Steel Institute (AISI) 1991] whereas AS4100 (SAA 1990) is quite conservative (Fig. 2).

Concrete Loaded CFTs

Existing Design Methods

The use of unbonded concrete filled steel tubes has been proposed by Orito et al. (1987). In this form of construction,

the axial load is applied directly to the concrete with the steel tube used only for confinement (no axial load). The authors found that the external steel tube improved the strength and the ductility of the internal concrete due to confinement.

Although this form of construction is relatively new, the unbonded circular steel tube achieves an effect similar to closely spaced circular ties in reinforced concrete, which can be modeled by either zero spacing between circular ties or triaxial compression tests. Attard and Setunge (1996) indicated that the behavior obtained from active and passive pressures are similar, whereas the load path only becomes significant for high confining pressures (ratio of confining stress to peak confined axial stress > 0.15).

Martinez et al. (1984) developed a simple model to predict the peak confined strength of high-strength concrete with circular ties. In comparison, Mander et al. (1988) developed a unified stress-strain model for normal- and high-strength concrete confined by either circular or square ties. The model proposed by Attard and Setunge (1996) can be used to predict the complete static response of the very-high-strength concrete with active confinement.

Experimental Verification

Circular thin-walled concrete filled steel tubes, with only the concrete loaded, have recently been extensively examined by O'Shea and Bridge (1997a,b). The parameters for the 18 test specimens are given in Table 1, including the peak axial load P and confining pressure p . The confining pressure was calculated using a pseudoelastic incremental confinement analysis (O'Shea and Bridge 1997a,b).

The median value for the ratio of test to predicted confined concrete strength using the models described previously has been included in Table 2. Two medians were calculated for each design method to allow a comparison over their calibrated range. Martinez et al. (1984) provided the best estimate of the peak confined concrete strength for nominal 50-MPa concrete, and Attard and Setunge (1996) provided the best estimate for 80- and 100-MPa concrete. Two design methods have been developed. The first can be used for unconfined concrete cylinder strengths ≤ 50 MPa, and the second can be used for unconfined cylinder strengths of 80 and 100 MPa. For values between 50 and 80 MPa, an average of both methods could be used.

Design Recommendations

In previous studies of concrete confinement (Martinez et al. 1984; Mander et al. 1988; Attard and Setunge 1996), several key parameters were used by the authors to calibrate their design model with their tests. These parameters include confined concrete strength f_{cc} , strain at peak confined concrete strength ϵ_{cc} , shape of the confined concrete curve, and confining pressure p . In the equation below, p and f_c are in megapascals.

Nominal 50-MPa Concrete. Although the equation proposed by Martinez et al. (1984) gives the best estimate of the confined concrete strength (Table 2), the complete stress-strain curve is not calculated. Consequently, the equation of Mander et al. (1988) has been used with new constants calibrated to the recent tests (O'Shea and Bridge 1997a,b). The modified equation is

$$f_{cc} = f_c \left(-1.228 + 2.172 \sqrt{1 + \frac{7.46p}{f_c}} - 2 \frac{p}{f_c} \right) \quad (1)$$

Nominal 80- and 100-MPa Concrete. For nominal 80- and 100-MPa concrete, the equations proposed by Attard and Setunge (1996) have been used

TABLE 1. Test Results for Concrete Loaded Specimens

Specimen (1)	<i>D</i> (mm) (2)	<i>t</i> (mm) (3)	<i>L</i> (mm) (4)	<i>f_c</i> (MPa) (5)	<i>E_c</i> (MPa) (6)	<i>f_y</i> (MPa) (7)	<i>E_s</i> (MPa) (8)	<i>p</i> (MPa) (9)	<i>P</i> (kN) (10)
S30CL50B	165.0	2.82	562.5	48.3	21,210	363.3	200,600	13.44	1,759
S30CL50C	165.0	2.82	571.0	38.2	16,140	363.3	200,600	12.45	1,649
S20CL50C	190.0	1.94	659.5	38.2	16,140	256.4	204,700	5.25	1,652
S16CL50B	190.0	1.52	658.0	48.3	21,210	306.1	207,400	4.86	1,841
S12CL50C	190.0	1.13	657.0	38.2	16,140	185.7	178,400	1.85	1,308
S10CL50C	190.0	0.86	657.5	38.2	16,140	210.7	177,000	1.52	1,240
S30CL80C	165.0	2.82	581.0	56.4	23,840	363.3	200,600	7.57	2,040
S20CL80C	190.0	1.94	655.5	56.4	23,840	256.4	204,700	4.39	2,338
S16CL80A	190.0	1.52	658.5	80.2	28,440	306.1	207,400	2.83	2,870
S12CL80C	190.0	1.13	661.5	56.4	23,840	185.7	178,400	1.21	1,862
S10CL80B	190.0	0.86	657.5	74.7	27,580	210.7	177,000	1.25	2,433
S10CL80C	190.0	0.86	664.5	56.4	23,840	210.7	177,000	1.26	1,940
S30CL10C	165.0	2.82	571.0	77.1	26,660	363.3	200,600	10.39	2,608
S20CL10C	190.0	1.94	656.0	77.1	26,660	256.4	204,700	3.82	3,083
S16CL10C	190.0	1.52	658.0	77.1	26,660	306.1	207,400	1.86	2,830
S12CL10C	190.0	1.13	662.5	77.1	26,660	185.7	178,400	0.64	2,630
S12CL10A	190.0	1.13	661.5	108.0	29,820	185.7	178,400	0.19	3,220
S10CL10C	190.0	0.86	664.0	77.1	26,660	210.7	170,000	0.30	2,553

TABLE 2. Comparison of Test/Predicted for Confined Concrete Strength and Models

Author (1)	Concrete Strength (MPa)	
	50 (2)	80 and 100 (3)
Martinez et al. (1984)	1.00	1.15
Mander et al. (1988)	0.92	1.06
Attard and Setunge (1996)	0.90	1.05

$$\frac{f_{cc}}{f_c} = \left(\frac{p}{f_t} + 1 \right)^k \quad (2)$$

$$k = 1.25 \left[1 + 0.062 \frac{p}{f_c} \right] (f_c)^{-0.21}; \quad f_t = 0.558 \sqrt{f_c} \quad (3a,b)$$

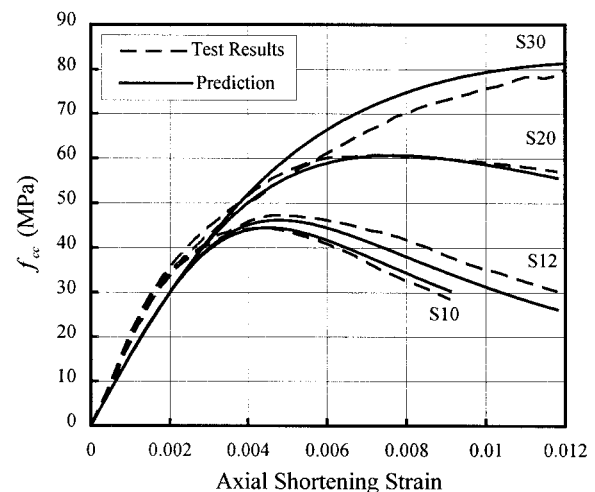
The strain at the peak confined concrete strength can be calculated using the equation proposed by Attard and Setunge (1996). A regression analysis using current tests [similar to that performed by Attard and Setunge (1996)] gave the following equation:

$$\epsilon_{cc} = \epsilon_c \left(1 + (8 + 0.05f_c) \left(\frac{p}{f_c} \right) \right) \quad (4)$$

Design Verification

The equations proposed by Mander et al. (1986) in conjunction with the modifications to peak confined concrete strength f_c [see (1)] and strain at the peak confined concrete strength ϵ_{cc} [see (4)] have been used to predict the response of the confined concrete with nominal unconfined cylinder strength of 50 MPa (Fig. 3). The equations provide a good estimate of the response of the confined concrete even in the postultimate region with the best estimates for low confining pressures such as specimens S10CL50C and S12CL50C (Table 1).

The equations proposed by Attard and Setunge (1996) in conjunction with modifications to the calculation of the strain at the peak confined concrete strength ϵ_{cc} [see (4)] have been used to predict the confined response of the high- and very-high-strength concrete (Fig. 4). A conservative estimate of the peak confined concrete strength and postultimate response was obtained. For specimen S12CL10C, the measured response reduced rapidly after the peak strength, largely due to the difficulty in testing under load control. For specimen S20CL10C,

**FIG. 3. Confined Concrete Response for Specimens Filled with Nominal 50-MPa Concrete Batch C**

the complete postultimate test response was unable to be obtained; however, the predicted response indicates the confined concrete curve.

An estimate of the confining pressure p is required to use the preceding equations. For a reinforced concrete design with circular ties, a common assumption is made that the circular ties are at full yield (Mander et al. 1988). However, O'Shea and Bridge (1997a,b) found that for thin-walled concrete filled steel tubes (loading only the concrete) this was not always valid. In Fig. 5, the confining pressures at maximum concrete enhancement have been normalized with the confining pressure assuming full yield of the steel tube ($p_{yield} = 2f_y t/D$) and plotted with the ratio of concrete to steel strength. A curve of best fit has been calculated with the equations shown in Fig. 5.

Concrete and Steel Loaded CFTs

Code Provisions

In O'Shea and Bridge (1997d), the current design models of Eurocode 4 (CEN 1992), ACI 318 (ACI 1989), and AS3600 (SAA 1994) have been compared to available test results on circular concrete filled steel tubes. The test results were restricted to the provisions in Eurocode 4 (CEN 1992) (i.e., concrete strengths not greater than 50 MPa and diameter-to-thick-

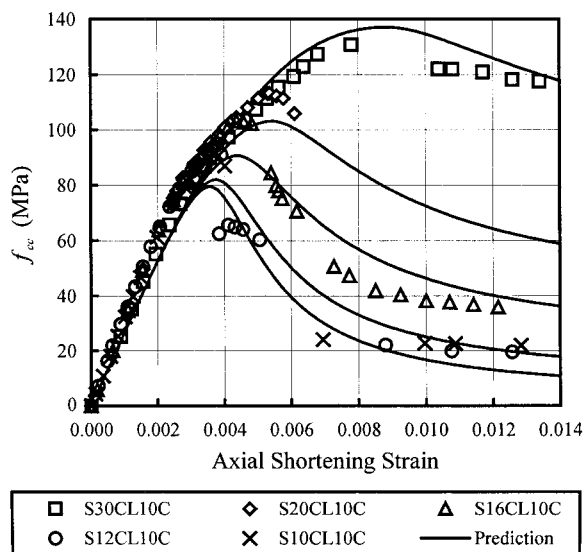


FIG. 4. Confined Concrete Response for Specimens Filled with Nominal 100-MPa Concrete Batch C

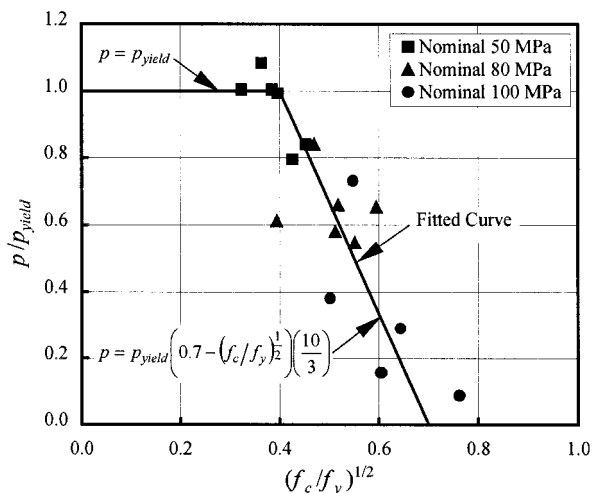


FIG. 5. Confining Pressure for Concrete Loaded Tests

ness ratios $<90(235/f_y)$, where f_y is the yield strength of the steel. The best estimate of strength was obtained from Eurocode 4 (CEN 1992) due to the inclusion of concrete enhancement from confinement and the use of a column curve for column slenderness.

Experimental Verification

The axial response of thin-walled concrete filled steel tubes with the concrete and steel loaded simultaneously has been experimentally investigated by O'Shea and Bridge (1997a,b). Three concrete mixes with nominal strengths of 50, 80, and 100 MPa were used to fill thin-walled steel tubes of varying diameter-to-thickness ratios. The dimensions of the specimens and their ultimate load P are given in Table 3.

The normalized response of the confined concrete in the tubes is given in Figs. 6 and 7. The response was obtained by subtracting the steel load component in the vertical direction from the applied load using an incremental analysis based on measured strains combined with a yield failure surface for the steel tube [as described in O'Shea and Bridge (1997a,b)].

In Figs. 6 and 7, for decreasing specimen D/t ratios, improvements in concrete strength and ductility can be seen. However, for specimens filled with very-high-strength concrete, the effect is marginal (ratio f_{cc}/f_c close to unity). Of note is the response of specimen S10CS80B (Fig. 7) with a higher ultimate load than equivalent thicker specimens (S12CS80A and S16CS80A). A local buckle was observed in specimen S10CS80B with a very small half-wavelength, similar to that observed by O'Shea and Bridge (1997c) for identical steel tubes with unbonded internal concrete restraint. Consequently, slip between the steel and the concrete must have occurred for specimen S10CS80B. Although large plastic deformations occurred in the steel tube in the locally buckled region, beyond this region the steel tube elastically unloads with a reduction in the vertical stress (compressive). This allowed greater circumferential stress (tensile) to be applied to the steel tube prior to the steel yield failure surface being reached, providing additional restraint to the internal concrete. As a result, the test was similar to the concrete loaded test with unbonded concrete with a corresponding higher load. This is in contrast to the results from specimens S16CS10A and S10CS10A. Although both of these specimens buckled, locally significant improvements in strength did not occur, indicating that little confinement can be achieved for very-high-strength concrete (unconfined strengths above 100 MPa).

Design Recommendations

Two design methods have been investigated in order to estimate the strength of axially loaded thin-walled steel tubes filled with concrete. The first method assumes that there is complete bond between the steel and concrete, and the second

TABLE 3. Test Results, Axially Loaded Thin-Walled Circular Concrete Filled Steel Tubes (CS)

Specimen (1)	D (mm) (2)	t (mm) (3)	L (mm) (4)	f_c (MPa) (5)	E_c (MPa) (6)	f_y (MPa) (7)	E_s (MPa) (8)	P (kN) (9)
S30CS50B	165.0	2.82	580.5	48.3	21,210	363.3	200,600	1,662
S20CS50A	190.0	1.94	663.5	41.0	17,810	256.4	204,700	1,678
S16CS50B	190.0	1.52	664.5	48.3	21,210	306.1	207,400	1,695
S12CS50A	190.0	1.13	664.5	41.0	17,810	185.7	178,400	1,377
S10CS50A	190.0	0.86	659.0	41.0	17,810	210.7	177,000	1,350
S30CS80A	165.0	2.82	580.5	80.2	28,450	363.3	200,600	2,295
S20CS80B	190.0	1.94	663.5	74.7	27,580	256.4	204,700	2,592
S16CS80A	190.0	1.52	663.5	80.2	28,450	306.1	207,400	2,602
S12CS80A	190.0	1.13	662.5	80.2	28,450	185.7	178,400	2,295
S10CS80B	190.0	0.86	663.5	74.7	27,580	210.7	177,000	2,451
S30CS10A	165.0	2.82	577.5	108.0	29,820	363.3	200,600	2,673
S20CS10A	190.0	1.94	660.0	108.0	29,820	256.4	204,700	3,360
S16CS10A	190.0	1.52	661.5	108.0	29,820	306.1	207,400	3,260
S12CS10A	190.0	1.13	660.0	108.0	29,820	185.7	178,400	3,058
S10CS10A	190.0	0.86	662.0	108.0	29,820	210.7	177,000	3,070

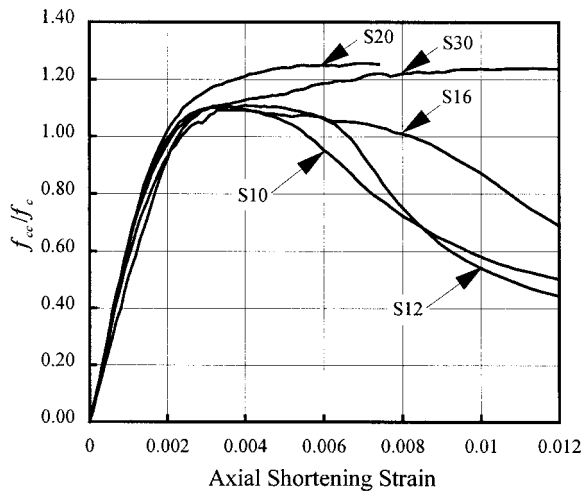


FIG. 6. Confined Concrete Response for Specimens Filled with Nominal 50-MPa Concrete

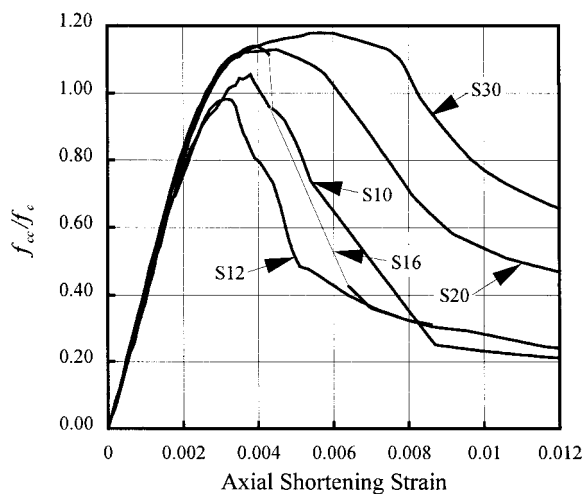


FIG. 7. Confined Concrete Response for Specimens Filled with Nominal 80-MPa Concrete.

method assumes that there is incomplete bond with local buckling of the steel tube.

Design with No Local Buckling. The test results from O'Shea and Bridge (1997a,b) have been compared to Eurocode 4 (CEN 1992) (denoted as EC4 in Table 4) with no reduction allowed for local buckling. In Eurocode 4 (CEN 1992)

the confinement of the concrete is a function of D/t , as well as the material properties. The steel reduction ratio f_v/f_y is the ratio of the axial steel strength to the calculated strength assuming full plasticity. The CS test refers to the concrete filled steel tube test (simultaneous loading of the concrete and steel) with the axial steel strength calculated from the measured strains on the steel tube (O'Shea and Bridge 1997a,b). The BS test is the locally buckled strength of the steel tube from a previously reported test series (O'Shea and Bridge 1997c). The column labeled EC4 is the calculated value from Eurocode 4 (CEN 1992) either for steel reduction or concrete enhancement.

Local buckling of the steel tube only occurred for a few specimens (S10CS80B, S16CS10A, and S10CS10A) with the steel reduction ratio f_v/f_y for the CS test significantly lower than the BS test. The local buckles occurred at the ends away from the strain gauges. For all other specimens where local buckling dominated the behavior of unfilled steel tubes (i.e., the thinner S10, S12, and S16 tubes), the steel reduction ratios for the CS tests are greater than the BS test. In Eurocode 4 (CEN 1992) the axial strength of the steel tube is reduced to account for hoop stresses from confinement of the internal concrete. This has been compared to the measured steel reduction ratio from the CS tests in Table 4. Eurocode 4 (CEN 1992) significantly overestimates the reduction to be applied to the steel for the current series of tests.

The observed (CS) and predicted (EC4) confined concrete strength f_{cc} are given in Table 4 normalized to the unconfined cylinder strength f_c . For specimens filled with nominal 50- and 80-MPa concrete, Eurocode 4 (CEN 1992) accurately predicts the confined concrete strength, the only exceptions being specimens S30CS50B and S30CS80A, where Eurocode 4 (CEN 1992) overestimates the confined concrete strength. As discussed in O'Shea and Bridge (1997c), this tube was produced by cold forming (compared with cold rolling for all of the other specimens) and had a rounded stress-strain curve with a high yield stress. Further analysis and possible experimental work needs to be conducted to evaluate the influence of these effects. Despite this, Eurocode 4 (CEN 1992) provides a good estimate of the ultimate strength with an average test to predicted a ratio of 0.99. However, for specimens filled with nominal 100-MPa concrete, Eurocode (CEN 1992) consistently overestimates the confined concrete strength.

For thin-walled steel tubes filled with very-high-strength concrete, Eurocode 4 (CEN 1992) is slightly unconservative (Table 4). A better estimate of strength can be obtained assuming no confinement of the internal concrete and consequently no reduction in the steel for biaxial effects.

TABLE 4. Comparison of Tests to Eurocode 4 (No Reduction for Local Buckling)

Specimen (1)	Steel Reduction f_v/f_y			Concrete Enhancement f_{cc}/f_c		CS Test/Eurocode 4 (CEN 1992)		
	CS (2)	BS (3)	EC4 (4)	CS (5)	EC4 (6)	Steel (7)	Concrete (8)	Composite (9)
S30CS50B	0.91	1.00	0.80	1.24	1.42	1.14	0.87	0.93
S20CS50A	0.94	0.97	0.80	1.26	1.21	1.18	1.04	1.06
S16CS50B	0.87	0.87	0.80	1.09	1.16	1.09	0.94	0.96
S12CS50A	0.94	0.88	0.80	1.11	1.08	1.17	1.02	1.04
S10CS50A	0.87	0.86	0.81	1.11	1.07	1.07	1.03	1.03
S30CS80A	0.92	1.00	0.81	1.18	1.24	1.14	0.95	0.96
S20CS80B	—	0.97	0.81	1.13	1.11	—	1.02	1.04
S16CS80A	—	0.87	0.81	1.06	1.10	—	0.97	0.99
S12CS80A	0.92	0.88	0.81	0.98	1.04	1.13	0.95	0.95
S10CS80B	0.73	0.86	0.82	1.14	1.04	0.90	1.10	1.09
S30CS10A	0.96	1.00	0.81	1.01	1.17	1.19	0.86	0.91
S20CS10A	1.00	0.97	0.82	1.04	1.07	1.22	0.97	1.00
S16CS10A	0.70	0.87	0.82	1.02	1.06	0.85	0.96	0.97
S12CS10A	0.96	0.88	0.82	0.98	1.03	1.17	0.95	0.96
S10CS10A	0.47	0.86	0.83	1.00	1.02	0.57	0.98	0.97

The provisions of Eurocode 4 (CEN 1992) can give conservative estimates of the axial strength of circular concrete filled steel tube for concrete strengths up to 80 MPa with the provisions for local buckling excluded (i.e., no account taken of local buckling for thin-walled steel tubes). For concrete strengths beyond 80 MPa, Eurocode 4 (CEN 1992) can be used to give a conservative estimate of the strength of circular concrete filled steel tubes with the provisions for confinement, steel reduction, and local buckling excluded (i.e., ignoring confinement and local buckling).

Design and Local Buckling. In this design model, local buckling of the steel tube is assumed to occur. Therefore, bond between the steel tube and the concrete infill cannot be maintained with slip occurring at the steel concrete interface. In recent tests on circular concrete filled steel tubes conducted by O'Shea and Bridge (1997a,b) this occurred in only three specimens. For concrete strengths up to and including 80 MPa, the cross section strength can be enhanced as greater circumferential restraint can occur. However, for specimens filled with nominal 100-MPa concrete, similar enhancement did not occur due to reduced circumferential expansion.

A design method has not been proposed for this model as insufficient tests were conducted to fully evaluate its effect. Instead, it is recommended that the thin-walled concrete filled steel tubes be conservatively designed according to the previous model, which assumes full bond between the steel tube and the internal concrete.

COMBINED COMPRESSION AND BENDING

Local Buckling of Circular Steel Tubes

Local buckling is significant for unfilled thin-walled circular steel tubes under uniaxial compression and combined compression and bending. Current design codes can provide a good estimate of the strength of the member in the region of high axial force and low moment (O'Shea and Bridge 1997c).

Tests performed on axially loaded circular steel tubes have shown that the buckling strength is unaffected by internal restraint (O'Shea and Bridge 1997c). Therefore, it is probable that this would also occur for members subjected to high axial force and low moment.

Circular Concrete Filled Steel Tubes

Code Provisions

In O'Shea and Bridge (1997d), three different design methods were compared to available test results: Eurocode 4 (CEN 1992); ACI 318 (ACI 1989); and AS3600 (SAA 1994). The Australian and American standards use a similar methodology that excludes the influence of concrete confinement by the steel tube and accounts for column slenderness by a minimum eccentricity. A more rational approach was adopted by Eurocode 4 (CEN 1992), which accounts for concrete confinement for certain specimen geometries, loading conditions, and column slenderness with a column curve. It was found that Eurocode 4 (CEN 1992) provided the best estimate of the specimens strength for axial compression and combined compression and bending. However, Eurocode 4 (CEN 1992) is only applicable for thick-walled steel tubes with diameter-to-thickness ratios <90 ($235/f_y$) and concrete strengths not greater than 50 MPa.

Experimental Verification

Numerous tests have recently been completed by O'Shea and Bridge (1997a,b) that examined the strength of short thin-walled steel tubes filled with medium- to very-high-strength concrete under combined compression and bending. The results have been fully described in O'Shea and Bridge (1997a,b), and important parameters are included in Table 5. It should be noted that thick end plates were used during the tests to ensure that the load applied at a constant eccentricity e resulted in a linear strain. Consequently, the actual column length is 81.5 mm longer than the specimen length L indicated in Table 5.

Design Recommendations

As Eurocode 4 (CEN 1992) is able to be successfully used (with some modifications) to reliably estimate the strength of axially loaded thin-walled steel tubes filled with concrete strengths up to 100 MPa, it will also be used as a basis for the design of circular thin-walled concrete filled steel tubes under combined compression and bending.

TABLE 5. Test Results, Eccentrically Loaded Thin-walled Circular Concrete Filled Steel Tubes (E1, E2)

Specimen (1)	D (mm) (2)	t (mm) (3)	L (mm) (4)	f_c (MPa) (5)	E_c (MPa) (6)	f_y (MPa) (7)	E_s (MPa) (8)	e (mm) (9)	P (kN) (10)
S30E250B	165.0	2.82	580.5	48.3	21,209	363.3	200,588	7.0	1,525
S20E250A	190.0	1.94	661.0	41.0	17,807	256.4	204,686	8.6	1,533
S12E250A	190.0	1.13	663.5	41.0	17,807	185.7	178,366	8.5	1,229
S10E250A	190.0	0.86	662.0	41.0	17,807	210.7	177,035	7.4	1,219
S30E150B	165.0	2.82	580.0	48.3	21,209	363.3	200,588	17.2	1,123
S20E150A	190.0	1.94	664.0	41.0	17,807	256.4	204,686	16.2	1,284
S16E150B	190.0	1.52	662.0	48.3	21,209	306.1	207,403	15.5	1,260
S12E150A	190.0	1.13	664.0	41.0	17,807	185.7	178,366	18.9	1,023
S10E150A	190.0	0.86	663.0	41.0	17,807	210.7	177,035	13.9	1,017
S30E280A	165.0	2.82	579.5	80.2	28,445	363.3	200,588	9.4	1,940
S20E280B	190.0	1.94	662.5	74.7	27,576	256.4	204,686	10.0	2,203
S10E280B	190.0	0.86	665.5	74.7	27,576	210.7	177,035	8.6	1,910
S30E180A	165.0	2.82	579.5	80.2	28,445	363.3	200,588	17.9	1,653
S20E180B	190.0	1.94	663.0	74.7	27,576	256.4	204,686	20.8	1,730
S16E180A	190.0	1.52	663.5	80.2	28,445	306.1	207,403	14.3	1,925
S10E180B	190.0	0.86	665.0	74.7	27,576	210.7	177,035	17.9	1,532
S30E210B	165.0	2.82	578.5	112.7	31,470	363.3	200,588	6.8	2,246
S20E210B	190.0	1.94	661.5	112.7	31,470	256.4	204,686	6.5	2,683
S10E210B	190.0	0.86	660.5	112.7	31,470	210.7	177,035	4.0	2,112
S30E110B	165.0	2.82	578.5	112.7	31,470	363.3	200,588	15.6	1,880
S20E110B	190.0	1.94	664.5	112.7	31,470	256.4	204,686	17.0	2,386
S16E110B	190.0	1.52	660.5	112.7	31,470	306.1	207,403	12.9	2,420
S12E110B	190.0	1.13	662.0	112.7	31,470	185.7	178,366	17.1	1,925

For the critical design case of axially loaded circular thin-walled concrete filled steel tubes, local buckling of the tube wall does not generally occur due to bond. It has been assumed that, for thin-walled circular concrete filled steel tubes with load applied at small eccentricities, local buckling of the tube wall will also not occur if full bond is maintained. Consequently, the provisions in Eurocode 4 (CEN 1992) can be used without alteration except for very-high-strength concrete where little concrete enhancement occurs.

The design method of Eurocode 4 (CEN 1992) uses an interaction diagram that can be calculated assuming full plasticity or more simply from a polygonal shape using specified points as indicated in Appendix C of Eurocode 4 (CEN 1992). Alternatively, a rigorous cross-sectional analysis can be performed.

The cross-sectional analysis was conducted using the fiber method (Wheeler and Bridge 1993) based on strain compatibility and material constitutive relationships. The uniaxial stress-strain curves for the materials were chosen for the analysis and have been fully described in O'Shea and Bridge (1997c). Two different concrete stress-strain curves were selected for each analysis.

In the first analysis, the unconfined dry concrete cylinder stress-strain curves were selected with the exception of the very-high-strength concrete. For this concrete, the unconfined dry cylinder response was modified to exclude strain reversal allowing the analysis to be performed over the complete strain range.

In the second analysis, the confined concrete curve [determined from uniaxial compression tests on concrete filled steel tubes (Figs. 6 and 7)] was used for the medium- and high-strength concrete. This analysis was not performed for the very-high-strength concrete as little concrete enhancement was observed in the uniaxial tests.

In Figs. 8–13, selected interaction curves using the rigorous cross-sectional analysis have been plotted together with Eurocode 4 (CEN 1992). Lines of constant eccentricity have been included at $e = D/20$ and $D/10$, and the test results are shown as solid black circles. The complete set of interaction curves can be found in O'Shea and Bridge (1997d).

For specimens filled with nominal 50-MPa concrete (Figs. 8 and 9), Eurocode 4 (CEN 1992) provides a reasonable estimate of the interaction curve. However, in regions of high axial force and low moment, Eurocode 4 (CEN 1992) can overestimate the specimens strength when compared to the fiber model using unconfined concrete. For specimens tested

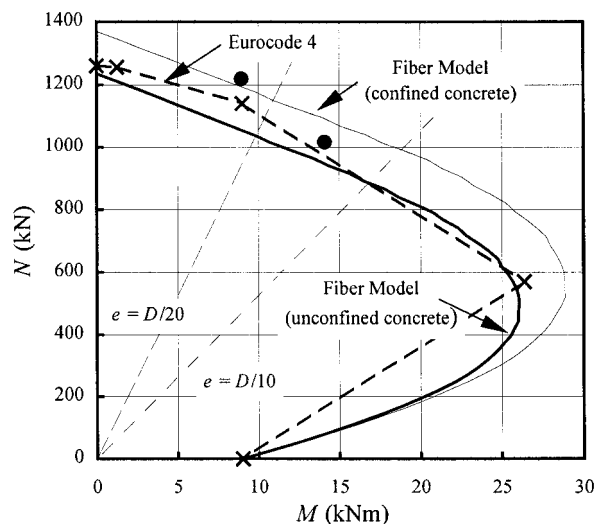


FIG. 8. Interaction Curve for S10 Tubes Filled with Nominal 50-MPa Concrete

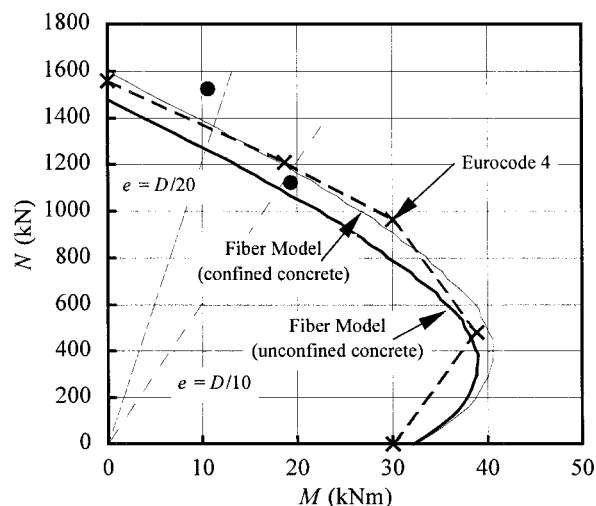


FIG. 9. Interaction Curve for S30 Tubes Filled with Nominal 50-MPa Concrete

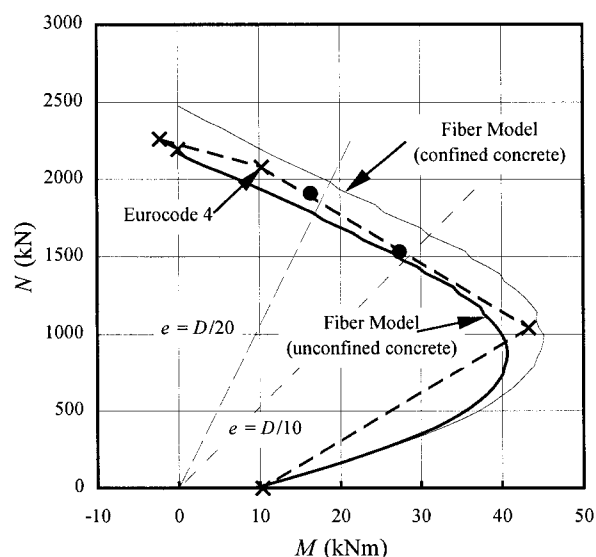


FIG. 10. Interaction Curve for S10 Tubes Filled with Nominal 80-MPa Concrete

at small eccentricities, the confined concrete interaction curve provides the best estimate of the strength, whereas the unconfined interaction curve is a better estimate at high eccentricities, suggesting that confinement of the concrete occurred at low eccentricities.

For specimens filled with nominal 80-MPa concrete (Figs. 10 and 11), the interaction curve calculated assuming full plasticity [Eurocode 4 (CEN 1992)] differs significantly from that obtained from the unconfined fiber model. However, as this is close to the confined interaction curve, Eurocode 4 (CEN 1992) indirectly provides a good estimate of the specimens strength. In regions of high moment, there is poor correlation between the unconfined fiber model and Eurocode 4 (CEN 1992). For the thinnest tube (S10) the simplified method described in Appendix C of Eurocode 4 (CEN 1992) gives an incorrect result (see point with a negative value of moment in Fig. 10).

Typical interaction curves calculated for specimens filled with nominal 100-MPa concrete are included in Figs. 12 and 13. There are significant differences in the results obtained from the full plastic analysis [Eurocode 4 (CEN 1992)] and the rigorous cross-sectional analysis (fiber model unconfined). A fiber analysis was conducted using fully plastic materials and has been included in Figs. 12 and 13 (fiber model 0.9M

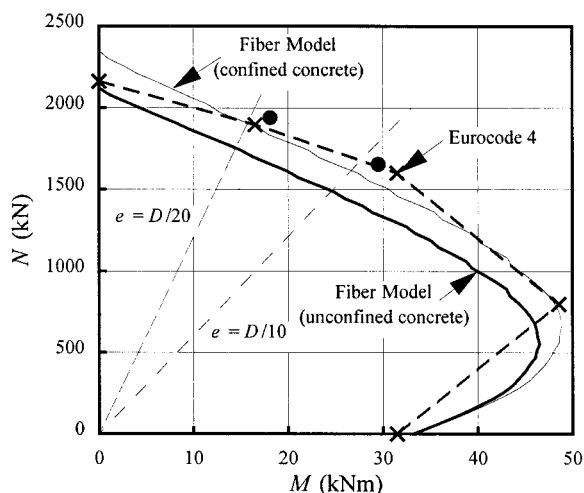


FIG. 11. Interaction Curve for S30 Tubes Filled with Nominal 80-MPa Concrete

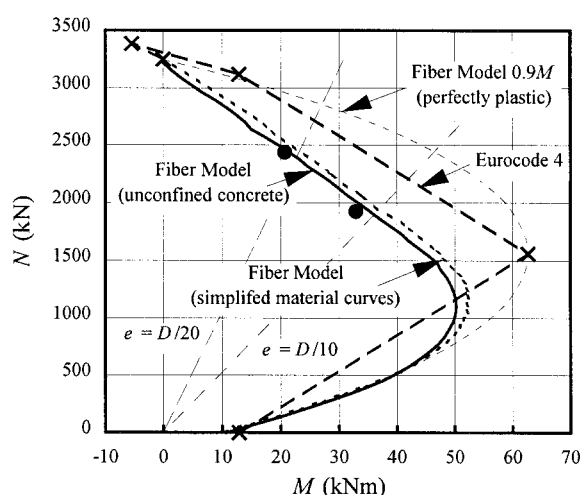


FIG. 12. Interaction Curve for S12 Tubes Filled with Nominal 100-MPa Concrete

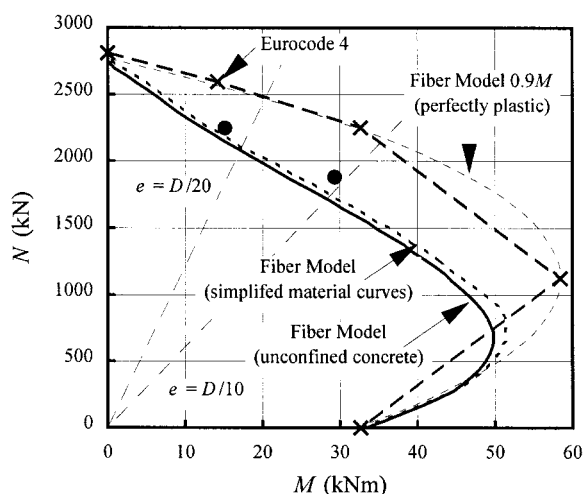


FIG. 13. Interaction Curve for S30 Tubes Filled with Nominal 100-MPa Concrete

perfectly plastic). Good correlation between the fiber model perfectly plastic and Eurocode 4 (CEN 1992) was obtained indicating that the equations are numerically correct. Consequently, although the equations proposed in Eurocode 4 (CEN 1992) are correct with the exception of one point (for very thin tubes) the assumption of full plasticity for very-high-

strength concrete is inappropriate. Using the material stress-strain curve proposed by Attard and Setunge (1996) and an elastic perfectly plastic curve for the steel, the fiber model labeled "Fibre model (simplified material curves)" has been included in Figs. 12 and 13. This differs slightly to the interaction curve obtained using the actual material properties, but is close, especially when compared to the perfectly plastic fibre model.

CONCLUSIONS

The strength of unfilled circular steel tubes have been found to be significantly affected by local buckling. Although the buckling strength of square tubes can be improved by providing internal lateral restraint, this was not observed in the circular steel tubes examined. Instead, the predominantly outward buckle remained unaffected by the internal concrete.

The degree of confinement offered by a thin-walled circular steel tube to the internal concrete is dependent upon the loading condition. The greatest concrete confinement occurs for axially loaded thin-walled steel tubes with only the concrete loaded and the steel tube used as pure circumferential restraint. Equations have been developed to predict the strength of this type of column and to provide an estimate of the shape of the postultimate curve.

For specimens within its calibrated range, Eurocode 4 (CEN 1992) has been shown to provide the best means for estimating the strength of circular concrete filled steel tubes with the concrete and steel loaded simultaneously. For axially loaded thin-walled steel tubes, local buckling of the steel tube does not occur if there is sufficient bond between the steel and concrete. For concrete strengths up to and including 80 MPa, the provisions of Eurocode 4 (CEN 1992) can be used with no reduction for local buckling. For concrete strengths in excess of 80 MPa, Eurocode 4 (CEN 1992) can still be used, but with no enhancement of the internal concrete from confinement and no reduction in the steel strength from local buckling and biaxial effects (from confinement). For the case in which local buckling of the steel tube occurs, an enhanced concrete strength can be obtained if the unconfined cylinder strength does not exceed 80 MPa. This is due to the steel tube unloading elastically in the region away from the local buckle allowing greater confinement of the concrete. This effect was not observed for specimens filled with high-strength concrete, as the concrete was unable to mobilize the additional confinement due to lower lateral expansion. Design rules have not been proposed for the local buckling case as insufficient tests have been performed. However, as local buckling results in a higher overall axial strength, a conservative estimate of the column strength can be made by assuming no local buckling.

Thin-walled circular steel tubes filled with medium strength concrete up to 50 MPa and subjected to combined axial compression and moment can be conservatively designed using the provisions in Eurocode 4 (CEN 1992) with no reduction for local buckling. For circular thin-walled steel tubes filled with high- and very-high-strength concrete up to 100 MPa and subjected to combined axial compression and moment, the interaction curve should not be calculated assuming full plasticity. Instead, the curve should be determined analytically using the actual material properties. For eccentricities less than $D/20$ and concrete strengths <80 MPa, a confined concrete stress-strain curve can be used (determined from uniaxial tests). In all other cases, the unconfined concrete stress-strain curve should be used.

Although the use of a rigorous cross-sectional analysis involves increased effort, the current simplified methods can give very unconservative strength estimates if used outside their calibrated range. Consequently, Eurocode 4 (CEN 1992) can only be used for the design of thin-walled steel tubes filled

with very-high-strength concrete if care is taken in the formulation of the design equations.

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APPENDIX I. REFERENCES

- American Concrete Institute (ACI). (1989). "Building code requirements for reinforced concrete." *ACI 318*, Detroit.
- American Iron and Steel Institute (AISI). (1991). *Cold formed steel design manual—load and resistance factor design*, Washington, D.C.
- Attard, M. M., and Setunge, S. (1996). "Stress-strain relationship of confined and unconfined concrete." *ACI Mat. J.*, 93(5), 432–442.
- Boué, P. (1957). "Concrete filled steel stanchions." *Acier Steel No. 9*, 351–356.
- Bridge, R. Q. (1990). "Design of composite columns." Postgraduate Course Notes, School of Civ. and Min. Engrg., University of Sydney, Sydney, Australia, 63–86.
- Cai, S. (1991). "Influence of moment distribution diagram on load-carrying capacity of CFST columns." *Proc., 3rd Int. Conf. on Steel-Concrete Compos. Struct.*, 113–118.
- Dowling, P. J., Janss, J., and Virdi, K. S. (1977). "The design of composite steel-concrete columns." *Proc., 3rd Int. Symp. of Stability of Steel Struct.: Introductory Rep., Appendix*, 309–327.
- European Committee for Standardization (CEN). (1992). "Design of composite steel and concrete structures, part 1.1—General rules and rules for buildings." *Eurocode 4*, Brussels.
- Furlong, R. W. (1967). "Strength of steel encased beam columns." *J. Struct. Div., ASCE*, 93(5), 113–124.
- Gardner, N. J. (1968). "Use of spiral welded steel tubes in pipe columns." *ACI J.*, 65(11), 937–942.
- Gardner, N. J., and Jacobson, E. R. (1967). "Structural behavior of concrete filled steel tubes." *ACI J.*, 64-38(July), 404–413.
- Grimault, J. P., and Janss, J. (1977). "Reduction of the bearing capacity of concrete filled hollow sections due to local buckling." *Proc., ECCS Colloquium on Stability of Steel Struct., Preliminary Rep.*, 175–179.
- Guiaux, P., and Janss, J. (1970). "Buckling behavior of columns made with steel tubes and filled with concrete." *CIDECT Issue 70/42/E*, No. MT 65.
- Imamura, T., et al. (1994). "Concrete filled steel tubular columns—Study and construction practice." *Proc., 4th ASCE Int. Conf. on Steel-Concrete Compos. Struct.*, 84–87.
- Kavoossi, H. R., and Schmidt, L. C. (1993). "The mechanical behaviour of higher strength concrete confined in circular hollow sections." *Proc., 13th Australasian Conf. on the Mech. of Struct. and Mat.*, University of Wollongong, Wollongong, Australia, 453–460.
- Kilpatrick, A. (1996). "The behaviour of high-strength composite concrete columns." School of Civ. Engrg., Curtin University of Technology, Australia.
- Klöppel K., and Goder, D. (1957). "Collapse load tests on concrete filled steel tubes and derivation of design formula." *Der Stahlbau*, Berlin.
- Knowles, R. B., and Park, R. (1969). "Strength of concrete filled steel tubular columns." *J. Struct. Div., ASCE*, 95(12), 2565–2587.
- Knowles, R. B., and Park, R. (1970). "Axial load design for concrete filled steel tubes." *J. Struct. Div., ASCE*, 96(10), 2125–2153.
- Lin, C. Y. (1988). "Axial capacity of concrete infilled cold-formed steel columns." *Proc., 9th Int. Spec. Conf. on Cold Formed Steel Struct.*, 443–456.
- Luksha, L. K., and Nesterovich, A. P. (1991). "Strength testing of large-diameter concrete filled steel tubular members." *Proc., 3rd Int. Conf. on Steel-Concrete Compos. Struct.*, 67–71.
- Mander, J. B., Priestley, M. J. N., and Park, R. (1988). "Theoretical stress-strain model for confined concrete." *J. Struct. Engrg., ASCE*, 114(8), 1804–1848.

- Martinez, S., Nilson, A. H., and Slate, F. O. (1984). "Spirally reinforced high-strength concrete columns." *ACI Struct. J.*, 81(5), 431–442.
- Masuo, K., Adachi, M., Kawabata, K., Kobayashi, M., and Konishi, M. (1991). "Buckling behaviour of concrete filled circular steel tubular columns using light weight concrete." *Proc., 3rd Int. Conf. on Steel-Concrete Compos. Struct.*, 95–100.
- Neogi, P. K., Sen, H. K., and Chapman, J. C. (1969). "Concrete-filled tubular steel columns under eccentric loading." *The Struct. Engrg., London*, 47(5), 187–195.
- Orito, Y., Sato, T., Tanaka, N., and Watanabe, Y. (1987). "Study on the unbonded steel tube composite system." *Proc., Compos. Constr. in Steel and Concrete*, ASCE, New York, 786–804.
- O'Shea, M. D., and Bridge, R. Q. (1997a). "Tests on circular thin-walled steel tubes filled with medium and high strength concrete." *Res. Rep. R755*, Dept. of Civ. Engrg., University of Sydney, Sydney, Australia.
- O'Shea, M. D., and Bridge, R. Q. (1997b). "Test on circular thin-walled steel tubes filled with very high strength concrete." *Res. Rep. R754*, Dept. of Civ. Engrg., University of Sydney, Sydney, Australia.
- O'Shea, M. D., and Bridge, R. Q. (1997c). "Local buckling of thin-walled circular steel sections with or without internal restraint." *Res. Rep. R740*, Dept. of Civ. Engrg., University of Sydney, Sydney, Australia.
- O'Shea, M. D., and Bridge, R. Q. (1997d). "Design of thin-walled concrete filled steel tubes." *Res. Rep. R758*, Dept. of Civ. Engrg., University of Sydney, Sydney, Australia.
- Prion, H. G. L., and Boehme, J. (1989). "Beam-column behaviour of steel tubes filled with high strength concrete." *Proc., 4th Int. Colloquium, North Am. Session on Code Differences Around the World*, Structural Stability Research Council, New York, 439–450.
- Rangan, B. V., and Joyce, M. (1992). "Strength of eccentrically loaded slender steel tubular columns filled with high-strength concrete." *ACI Struct. J.*, 89(6), 676–681.
- Roik, E. K., and Bergmann, R. (1989). "Harmonisation of the European construction codes, report on Eurocode 4." *Rep. EC4/6/89*, Ruhr University, Bochum, Germany.
- Sakino, K., and Hayashi, H. (1991). "Behaviour of concrete filled steel tubular stub columns under concentric loading." *Proc., 3rd Int. Conf. on Steel-Concrete Compos. Struct.*, 25–30.
- Salani, H. J., and Sims, J. R. (1964). "Behaviour of mortar filled steel tubes in compression." *ACI J.*, 61(10), 1271–1283.
- Sen, H. K. (1972). "Concrete filled tubular steel columns." *Tubular Structures 17*, Tubes Division, British Steel.
- "Steel structures." (1990). *AS4100*, Sydney, Australia.
- "Concrete Structures." (1994). *AS3600*, Sydney, Australia.
- University of Sydney School of Civil Engineering. (1990). "New commonwealth offices alternative composite column design." *Investigation Rep. No. S837*, Ctr. for Advanced Struct. Engrg., University of Sydney School of Civil and Mining Engineering, Sydney, Australia.
- Wheeler, A. T., and Bridge, R. Q. (1993). "Analysis of cross-sections in composite materials." *Proc., 13th Australian Conf. on the Mech. of Struct. and Mat.*, 929–936.

APPENDIX II. NOTATION

The following symbols are used in this paper:

- A_s = cross-sectional area of steel;
 D = tube diameter;
 E_c = elastic modulus for concrete;
 E_s = elastic modulus for steel;
 e = eccentricity of loading;
 f_c = unconfined concrete cylinder strength;
 f_{cc} = peak confined concrete strength;
 f_t = tensile concrete strength;
 f_v = axial steel strength;
 f_y = steel strength;
 k = confinement coefficient;
 L = specimen length;
 N = axial force;
 M = moment;
 P = peak load;
 p = applied confining pressure;
 p_{yield} = confining pressure assuming steel tube at full yield;
 t = wall thickness;
 ϵ_c = strain at peak unconfined concrete strength; and
 ϵ_{cc} = strain at peak confined concrete strength.